MLE-AI-201 · FINAL PROJECT · AI ACADEMY COHORT II

PegaSold

Baku property prices, predicted by a decision tree, a Pegasos SVM and a bagging ensemble, all written from scratch in NumPy.



57,003	82	0.886	93.4%
listings, cleaned	features	R^2 on log price	tier accuracy

Aygun Asvarova · Emiliya Abdulhalimova · Kanan Karimli · Murad Bagirzade

THE PROBLEM

Two tasks on one dataset

Task A · regression

log(price)

Predict the asking price of a listing.

Task B · classification

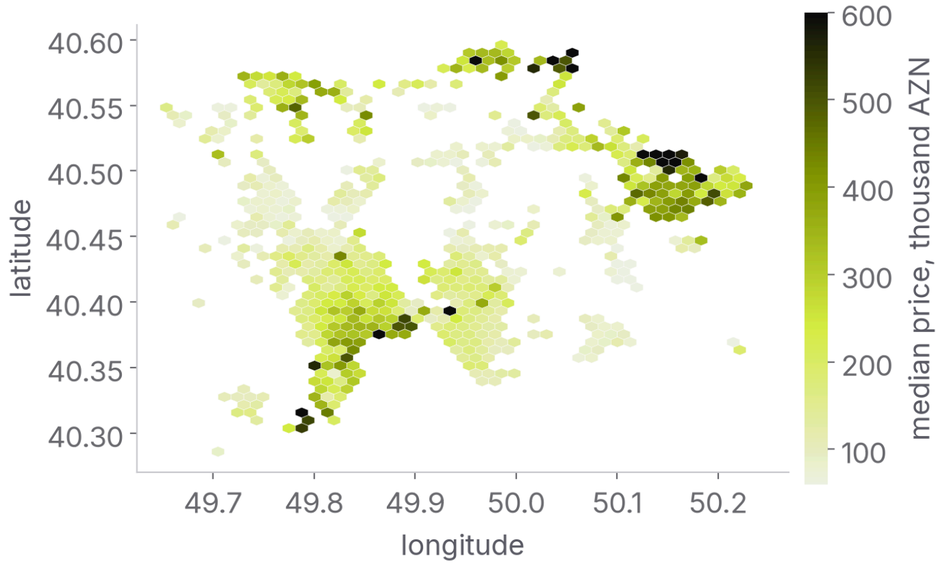
209,000 AZN

Premium or standard, split at the median of the training set.

The rule

The tree and the SVM had to be our own NumPy. scikit-learn appears only as a baseline.

Median advert price per hexagon



The price surface is a bowl, not a plane: expensive by the bay, fading outwards.

EXPLORATION

Four ways the raw file misleads you

36%

repeat scrapes

100,775 rows, only 64,454 adverts. One appears 17 times.

3

columns that leak

total_price is the target renamed; unit_price × area rebuilds it; description states it.

91.5%

“missing” VIP

One distinct value. Empty means no, not unknown.

100×

two units, one column

Area is m², except for land, where it is sot.

We also refuse *views*

It is only recorded after the advert goes live, so it cannot be known about a flat you are about to price.

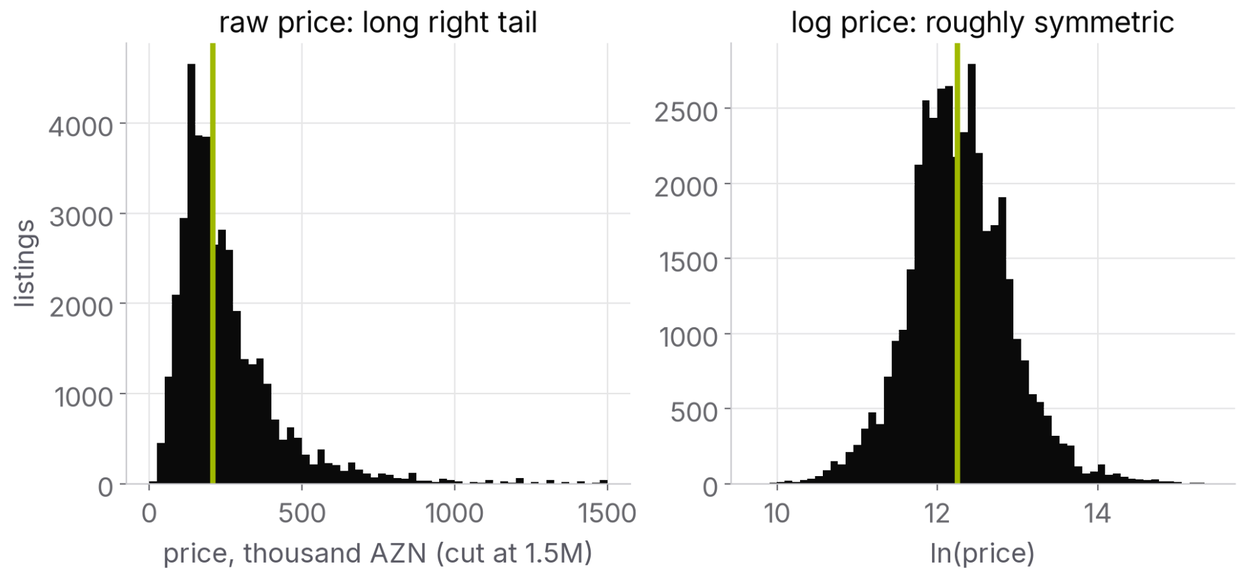
Why it matters

Every one of these is silent: no error is raised, the model simply learns something false.

THE TARGET

Why we model log(price)

Cleaned training prices: raw vs log



342k vs 218k

mean vs median, raw file

Six orders of magnitude, from 11 AZN to 600M.

0.65

corr(log area, log price)

Close to a straight line: the relationship becomes additive.

PREPARATION

100,775 rows → 57,003

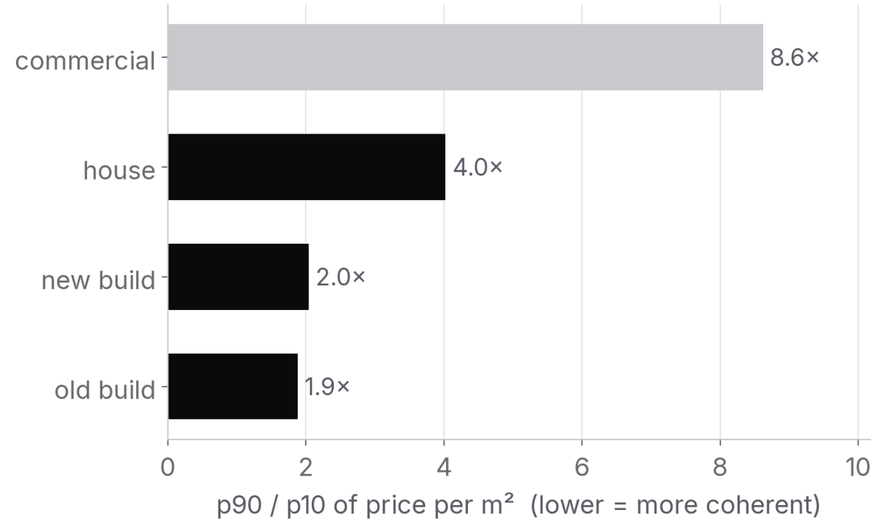
Raw file	100,775
Repeat scrapes	−36,990
Reposted twins	−544
Scope: Baku residential	−6,172
Impossible values	−66 → 57,003

last sighting kept: 78% of price moves are down

deduplicated before the split

removed, never clipped

How coherent is each category's pricing?



Land has no floor and no rooms. These are not gaps to impute: the question does not apply.

PROTOCOL

The test set decided nothing

Learned on train only

fill medians | scaler μ, σ | location list | tier threshold

Median of the whole file: 218,000. Median of the training split: 209,000. That gap is the leak we did not take.

61 / 61

automated checks pass

No advert in two splits. Labels match the training threshold. Leaking columns absent.

70/15/15

split, seed 42

Hashed listing IDs per split, so we can prove it never moved.

Reproducible

`python -m src.run_all` rebuilds every number and figure in this deck in a few minutes.

FEATURES

19 working columns → 82 features

11 continuous	9 flags	3 type	59 location one-hots

Engineered, not copied

distance_to_centre_km · floor_ratio · is_top_floor · area_per_room · has_floor_info

Distance turns two coordinates into the one measurement a linear model can use.

Scaled on train

Area in the hundreds cannot share a weight vector with a 0/1 flag. The SVM needs standardisation; our tree is indifferent to it.

has_floor_info exists so the model can tell a real zero from a value we filled in.

The decision tree

$$\text{gain} = \mathrm{I}(\text{parent}) - (n_L/n)\cdot\mathrm{I}(\text{left}) - (n_R/n)\cdot\mathrm{I}(\text{right})$$

Criterion

Gini or entropy for classification, variance for regression. Leaves predict the majority class or the mean.

Split search

Sort each column once; one cumulative sum scores every candidate threshold as a vector. Quadratic becomes one sort.

Guards

Depth cap, pure node, no positive gain, and leaves of 5: a one-row leaf claims probability 1.0 and ranks badly.

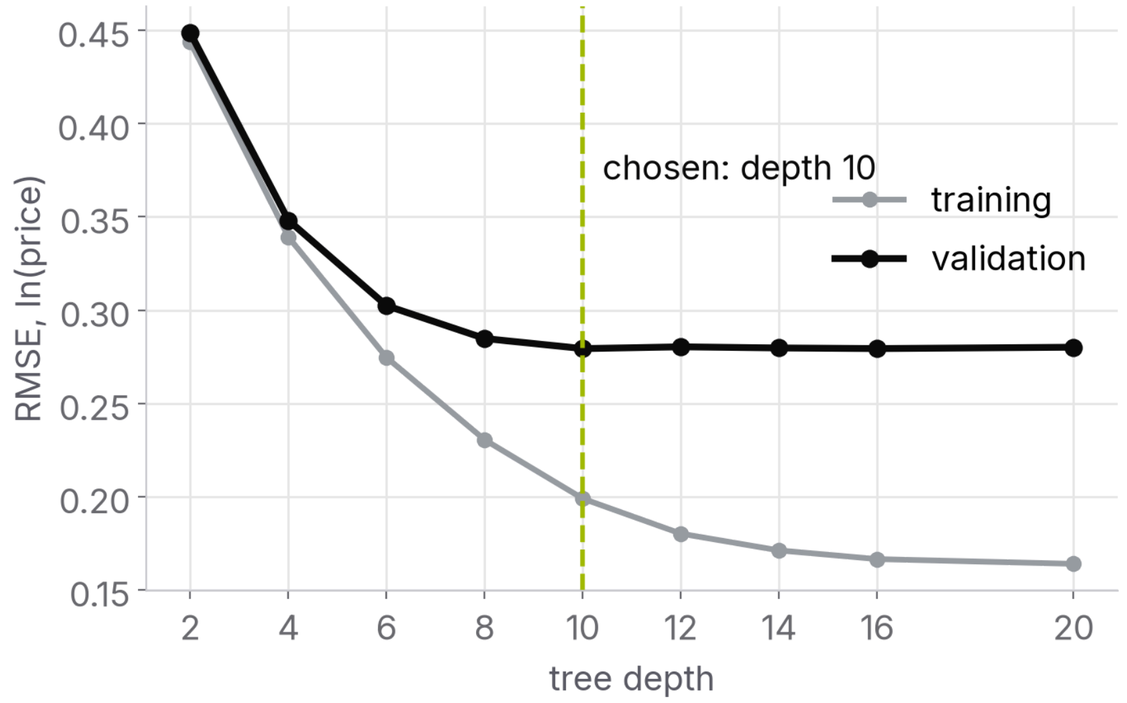
Numerical detail

Log prices sit around 12, so we centre the target before the prefix sums. Gains are translation-invariant: identical in exact arithmetic, stable in floating point.

MODEL SELECTION

Where the tree starts memorising

Error vs tree depth, train and validation (price)



10

depth, regression

8

depth, classification

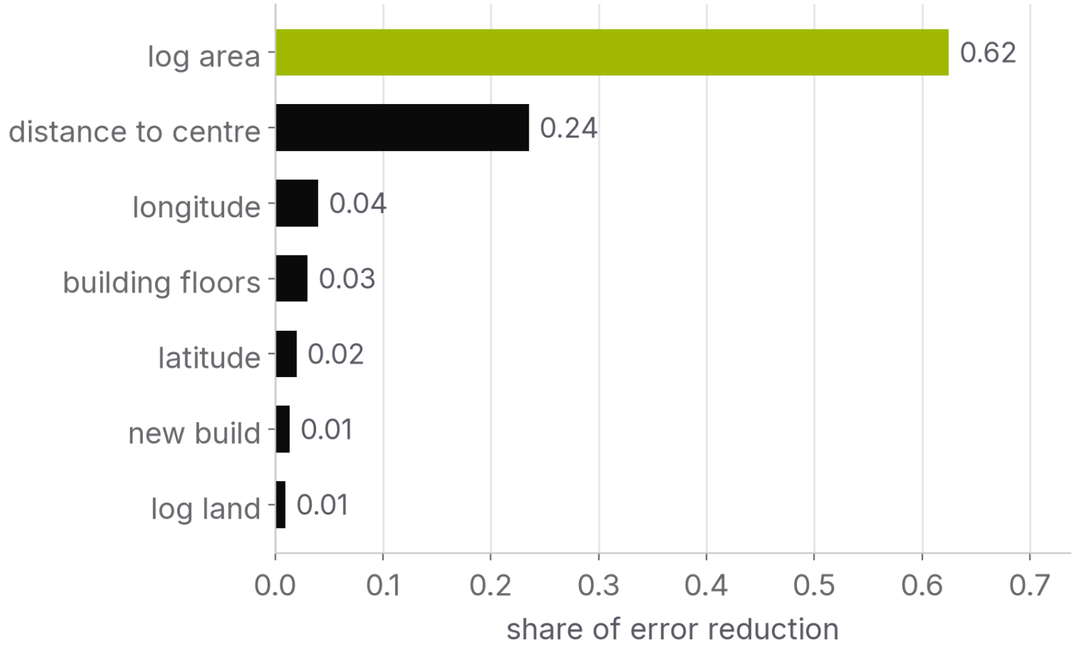
Chosen on validation

Train error keeps falling; validation flattens, then turns. We stop at the turn.

INTERPRETATION

What the tree actually used

Regression tree feature importance



0.62

log area

0.24

distance to centre

Nothing else reaches 0.05.

Read it carefully

Weighted impurity decrease describes how this tree split, not causation. Area and rooms correlate at 0.82 and can trade credit.

MODEL 2 · FROM SCRATCH

Thirty of our trees beat one of our trees

30
trees

6,000
bootstrap rows each

16
depth per member

−26%
validation MSE

Deeper members, on purpose

The sweep asks when one tree overfits: the wrong question once you average. Depth 8 → 0.914; depth 16 → 0.928.

Bagging, not a random forest

Subsampling features scored worse: 0.0429 against 0.0421 MSE. 59 of 82 features are sparse location one-hots.

Out-of-bag check

Scoring each row with only the trees that never saw it: R^2 0.888, accuracy 0.929, matching validation with no validation data.

MODEL 3 · FROM SCRATCH

The Pegasos SVM

$$\min \ (\lambda/2)\|w\|^2 + (1/n)\,\Sigma\,\max(0,\,1 - y_i(w\cdot x_i + b)) \quad \cdot \quad \eta_t = 1/(\lambda t)$$

The projection is not optional

With λ = 0.001 the first learning rate is 1000. Without projecting w back into the ball, the first updates leave the feasible region and the model depends on the shuffle order.

The intercept needed a cap

It is deliberately unregularised, but at those early rates an uncapped update inflates the bias on centred data. We cap its step at min(η, 1).

0.001
λ, chosen on validation

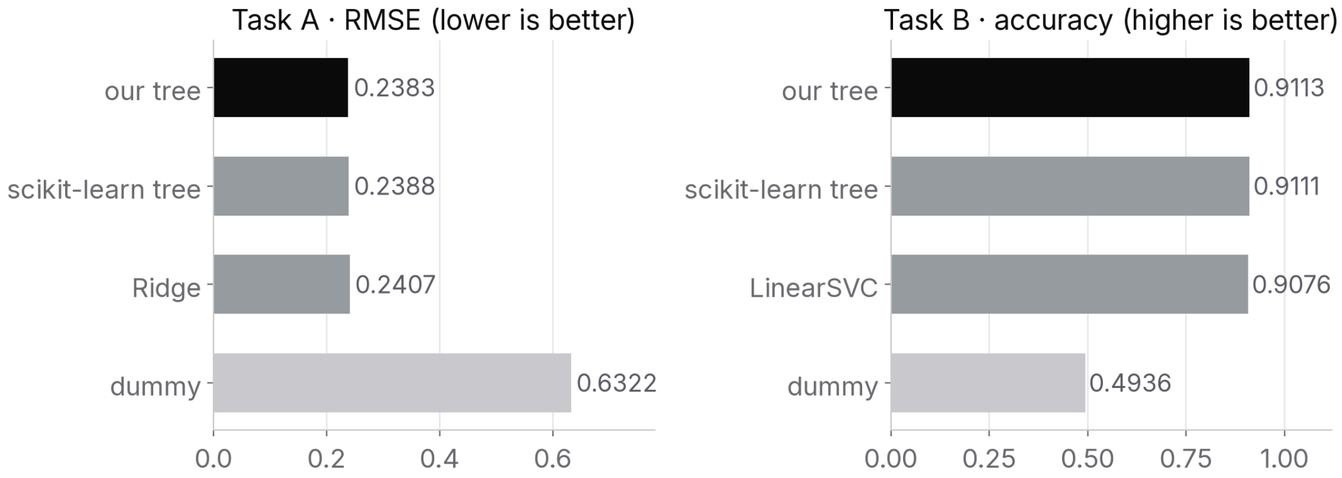
11,077
support vectors

0.842
RBF-feature variant: worse than linear 0.908

VERIFICATION

Does our code actually work?

Validation, hyper-parameters matched



Agreement is the result

Agreement to three decimals says our splitting logic is correct, not that we out-engineered a mature library.

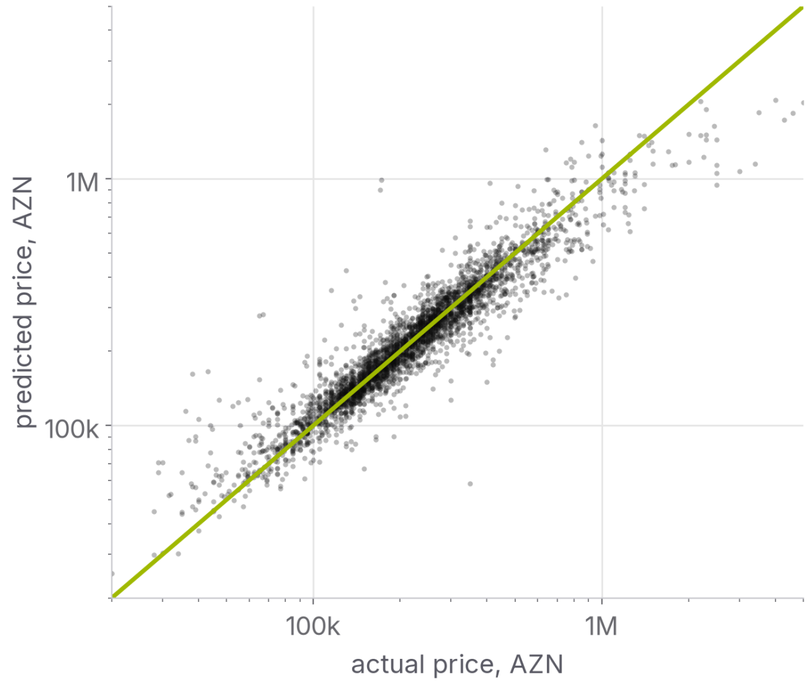
Matched hyper-parameters

RMSE 0.2383 vs 0.2388. Accuracy 0.9113 vs 0.9111. Five-fold CV agrees too.

RESULTS · TASK A

Price

Bagged trees: predicted vs actual (test)



TEST SET	TREE	BAGGED
R^2	0.856	**0.886**
RMSE (log)	0.245	**0.218**
MAE (log)	0.164	**0.139**
Typical error	17.8%	**14.9%**

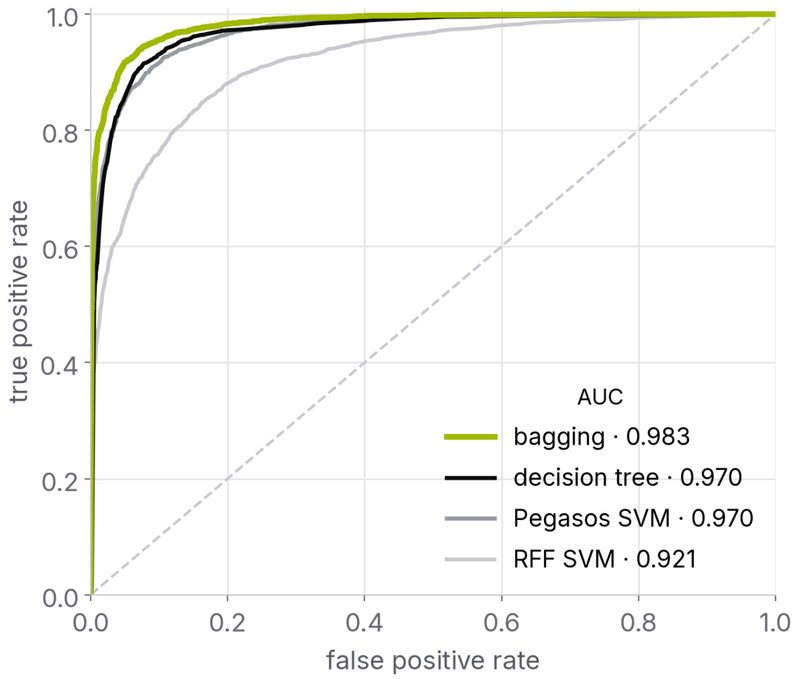
Averaging alone

Bought 26% of the squared error on validation. No new features, no new information.

RESULTS · TASK B

Price tier

ROC curves on the test set



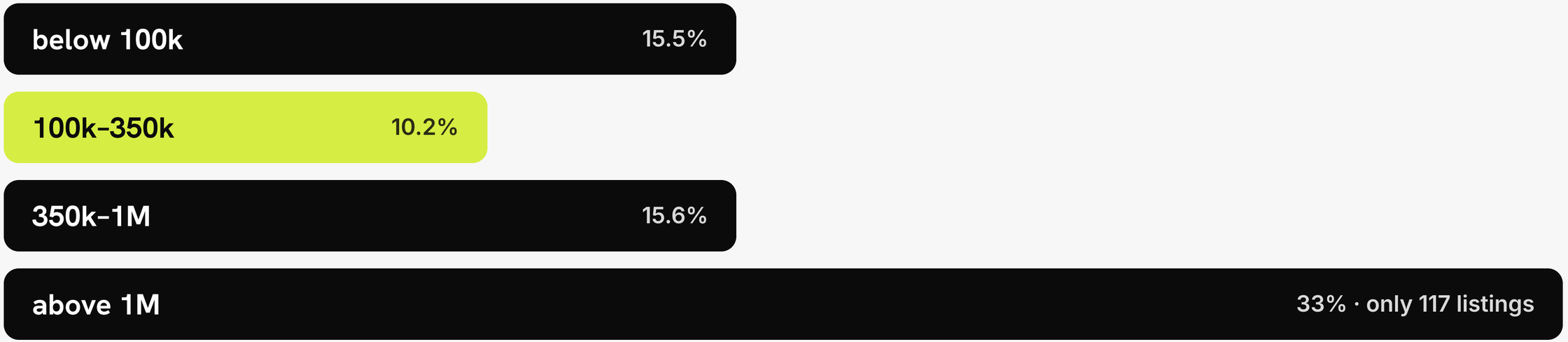
TEST SET	ACC	AUC
Our DecisionTree	0.9159	0.9695
Our PegasosSVM	0.9084	0.9698
Our BaggedTrees	**0.9337**	**0.9827**
Dummy	0.4936	—

Same AUC, different accuracy

Tree and SVM rank listings equally well. They differ only in where one straight cut can land.

HONESTY

Where the models fail



Median |error|, single tree on validation.

What the data is

Asking prices, not sales, from a 45-day snapshot in autumn 2024.

What we left out

Baku residential only, by design; the description text is dropped because it contains the price.

CONTRIBUTIONS

Who did what, and what comes next

Emiliya Abdulhalimova

Exploratory analysis and the cleaning pipeline: explore.py, data_prep.py, scope and filter decisions.

Next

Target-encode location, cross-fitted.

Aygun Asvarova

Features, the train-only preprocessor, the split and its fingerprints, model selection sweeps.

Next

Strip prices from the description text.

Kanan Karimli

decision_tree.py, the bagging ensemble, tree visualisation and the web demo.

Next

Histogram split search → all 39,902 rows.

Murad Bagirzade

svm.py and its nonlinear variant, validate.py and its 61 checks, evaluation.

Next

A separate model for the luxury tail.

IN CLOSING

What we would defend hardest

The result

Two models and an ensemble, written from scratch on 57,003 cleaned listings.

0.886 R^2

93.4% · 0.983

But the part that makes it mean anything

- **37,534 duplicates removed before the split**
- **3 leaking columns refused**
- **every statistic learned on train**
- **the test set scored once**

Any one of those, left alone, would have produced a better-looking number and a worse model.

Thank you. Questions?

LIVE DEMO

Pick a home. Get a price.

Input

Home type, area, location, rooms, floor and a few yes/no features.

Output

Price, premium / standard tier and the model's confidence.

Example

New build, Nərimanov, 85 m², 3 rooms → 248,361 AZN, Premium.

```
python -m src.web.web_demo
```

